

Introduction to Morphological Image Processing

UNIT 5 | NCCS

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Image Processing | CSIT 5

Unit 5	Introduction to Morphological Image Processing	Teaching Hours (2)
Introduction	Logic Operations involving binary images, Introduction to Morphological Image Processing, Definition of Fit and Hit	1 hr
Morphological Operations	Dilation and Erosion, Opening and Closing	1 hr

Morphological Image Processing

Morphological Image Processing is a set of image processing techniques based on shape and structure of objects in an image.

- *It works mainly on binary images (also grayscale).*
- *Uses a small pattern called a Structuring Element (SE).*
- *Operations probe the image to extract or modify object shapes.*

Common Morphological Operations

- Erosion
- Dilation
- Opening
- Closing
- Hit-and-Miss Transform

👉 In simple words:

Morphology = shape-based image processing

Basic idea

“How does an object look? How big is it? Where are its boundaries?”

1 HIT Condition

✓ HIT means match condition is satisfied.

In simple words:

If the structuring element (SE) fits the object pixels (1's) exactly → it is a HIT.

If we place the SE on the top-left corner:

sql

1 1 ← matches
1 1

All 1's match exactly ✓

👉 This is a **HIT**

Example:

Suppose we have this binary image:

1 1 0
1 1 0
0 0 0

And structuring element:

1 1
1 1

2 MISS Condition

✗ MISS means condition is not satisfied.

If the structuring element does NOT match the object pixels → it is MISS.

Example:

If we place the same SE at wrong position:

```
1 0
1 0
```

It does not match

👉 This is a **MISS**

Dilation and Erosion

In image processing (especially Morphological Image Processing), Dilation and Erosion are two basic operations used mainly on binary images (also grayscale images) to modify object shapes using a small shape called a structuring element.

☞ Erosion (Shrinking / Thinning)

Erosion removes pixels from the boundaries of objects. So the object becomes smaller or thinner.

Think of it like: 🗑️ Soil erosion — edges get worn away

Rule:

If all 1's of the structuring element fit inside object pixels → output = 1

So in erosion:

- ✓ All SE 1's match image 1's → result = 1
- ✗ Even one mismatch → result = 0

In EROSION

This is like an **AND condition**

So:

Erosion = "All must match → shrink"

🌸 Dilation (Growing / Thickening)

*Dilation adds pixels to the boundaries of objects. So the object becomes bigger or thicker.
Think of it like: 🌱 Plant growth — object expands 🌿*

Rule:

If **at least one** pixel of the structuring element (SE) overlaps a 1 → output = 1

So in dilation:

- ✓ Any one match → result is 1
- ✗ No match → result is 0

In DILATION

This is like an **OR condition**

So you can think:

Dilation = "If any 1 touches → grow"



Original Binary Image

Consider this small image:

```

0 0 0 0 0
0 1 1 0 0
0 1 1 0 0
0 0 0 0 0

```

This is a small **2×2 white square** in the center.

Structuring Element (SE):

Use a simple **3×3 square**

```

1 1 1
1 1 1
1 1 1

```

**1**

DILATION

Rule:

If any 1 of SE overlaps a 1 in image → output = 1

What Happens?

The white region **expands (grows)**.

Result After Dilation:

```

0 1 1 1 0
1 1 1 1 0
1 1 1 1 0
0 1 1 1 0

```



The object becomes bigger.



Why Did It Disappear?

Because erosion requires:

ALL pixels of the 3×3 structuring element must match.

But our object is only 2×2 → too small.

**2**

EROSION

Rule:

All SE 1's must fit inside object pixels.

Since our object is small (2×2),
the 3×3 SE **cannot fully fit inside it**.

Result After Erosion:

```

0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0

```



Object disappears completely.

0	0	0	0	0
0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

Segment A

Image segment

0	1	0
1	1	1
0	1	0

Segment B

Structuring element

$\Rightarrow$ Erosion ($A \ominus B$)

0	0	0	0	0
0	0	0	0	0
0	0	1	0	0
0	0	0	0	0
0	0	0	0	0

$\Rightarrow$ Dilation ($A \oplus B$)

0	1	1	1	0
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

Practical Applications of Dilation and Erosion:

1. Noise Removal:

- Use erosion to remove small noise points, followed by dilation to restore the size of objects.

2. Gap Filling:

- Use dilation to fill small gaps in objects.

3. Object Separation:

- Use erosion to separate connected objects.

4. Boundary Detection:

- Subtract the eroded image from the original image to detect object boundaries.

Summary:

- **Dilation:** Expands white regions.
- **Erosion:** Shrinks white regions.

Example in Real-World Image Processing:

Suppose you have a binary image of a handwritten letter with small gaps and noise:

- Apply **dilation** to fill the gaps and connect broken parts.
- Apply **erosion** to remove noise and thin out the letter.

By combining these operations (e.g., opening = erosion followed by dilation, closing = dilation followed by erosion), you can achieve more advanced image processing tasks.

Both operations rely on a structuring element and are fundamental to morphological image processing. They are widely used in applications like medical imaging, object detection, and document analysis.

Explain Opening and closing morphological operations in brief

- Opening = Erosion → then Dilation
- Closing = Dilation → then Erosion



Original Image

```
0 0 0 0 0 0
0 0 1 1 0 0
0 1 1 1 1 0
0 0 1 1 0 0
0 0 0 0 0 0
```

Structuring Element (SE): 3×3 square

```
1 1 1
1 1 1
1 1 1
```



This is a small object with a **small noise pixel** attached.

1 OPENING (Erosion → Dilation)

Step 1: Erosion

Small thin parts or noise disappear.

After erosion:

```
0 0 0 0 0 0
0 0 0 0 0 0
0 0 1 1 0 0
0 0 0 0 0 0
0 0 0 0 0 0
```

Small extra parts are removed.

Step 2: Dilation

The remaining part grows back slightly.

Final Opening Result:

```
0 0 0 0 0 0
0 0 1 1 1 0
0 0 1 1 1 0
0 0 1 1 1 0
0 0 0 0 0 0
```

- ✓ Small noise removed
- ✓ Object becomes smoother
- 👉 Opening removes small objects.

2 CLOSING (Dilation → Erosion)

Now start from original image.

After Dilation:

```
0 1 1 1 1 0
1 1 1 1 1 1
1 1 1 1 1 1
1 1 1 1 1 1
0 1 1 1 1 0
```

Now the object is much bigger (expanded boundary).

After Erosion:

```
0 0 0 0 0 0
0 1 1 1 1 0
0 1 1 1 1 0
0 1 1 1 1 0
0 0 0 0 0 0
```

- Shape becomes smoother
- Small holes would be filled
- Object is slightly thicker than original

Opening:-

- Erosion is followed by dilation operation.

$$A \circ B = (A \ominus B) \oplus B$$

Original Image $\swarrow$ $\downarrow$ Structural Image.

- Identify gaps in an Image
- Edges become sharp (or) smooth
- Isolates objects which are touching one another.

Closing:

— Dilation is followed by Erosion

$$A \bullet B = (A \oplus B) \ominus B$$

Original
Image

Structural Image.

— used to fuse narrow breaks
& eliminate small holes

Ex:-

(A)

1	1	1	1	1	1	1
1	1	1	0	1	1	1
1	1	1	0	1	1	1

Original Image

(B)

1	1	1
1	1	1
1	1	1

Structural
Image

Eg-

0	0	0	0	0
0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

Segment A

0	1	0
1	1	1
0	1	0

Segment B

Closing $(A \bullet B)$

$$A \bullet B = (A \oplus B) \ominus B$$

$$= \begin{pmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{pmatrix} \ominus \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Opening $(A \bullet B)$

$$A \bullet B = (A \ominus B) \oplus B$$

$$= \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Define region and Boundary.

- 1.Region** : In image processing, a **region** refers to a connected set of pixels within an image that share similar properties or characteristics. These properties could be color, intensity, texture, or any other feature that distinguishes the region from its surroundings. Regions are often used in segmentation tasks to group pixels into meaningful objects or parts of an image.
- 2.Boundary** : A **boundary** (or edge) in image processing is the line or contour that separates one region from another. It represents the transition between different regions and is typically characterized by significant changes in pixel intensity, color, or texture. Boundaries are crucial for object detection, shape analysis, and image segmentation.

List some elements of Visual Perception

Visual perception refers to the process by which humans interpret and organize visual information from the environment. In the context of image processing, understanding visual perception helps in designing algorithms that mimic human vision. Some key elements of visual perception include:

1. Brightness :

- Refers to the perceived intensity of light. In image processing, brightness can be adjusted to enhance or reduce the visibility of certain features in an image.

2. Contrast :

- Contrast is the difference in brightness or color between objects or regions in an image. High contrast makes objects stand out, while low contrast can make them blend together. Contrast enhancement is a common operation in image processing.

3. Color :

- Color perception is based on the wavelength of light reflected or emitted by objects. In image processing, color spaces like RGB, HSV, or CMYK are used to represent and manipulate colors.

4. Texture :

- Texture refers to the arrangement of pixels in a region that gives the perception of surface properties, such as roughness or smoothness. Texture analysis is used in image segmentation, classification, and object recognition.

5. Shape :

- Shape perception involves recognizing the geometric structure of objects in an image. Shapes can be described using contours, boundaries, or skeletons, and are important for object recognition and classification.

6. Edges and Contours :

- Edges are sharp changes in intensity or color that define the boundaries of objects. Contours are continuous lines that outline the shape of an object. Edge detection algorithms (e.g., Canny edge detector) are fundamental in image processing.

2. Write short notes on dilation and erosion.

10. Explain opening and closing morphological operations in brief.

Define region and Boundary. List some elements of Visual Perception [2+3]